SOFTWARE ENGINEERING LAB – LAB 1
Requirements Engineering & UML Use-Case Modelling
Hospital Bed & ICU Allocation Dashboard

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Problem Statement #20

Hospital Bed & ICU Allocation Dashboard

An emergency room bed management dashboard displaying real-time ward occupancy, prioritizing ICU transfers based on triage score, and orchestrating inter-facility ambulance transfer workflows.

Target Stakeholders / Actors:

Triage Nurse

Hospital Administrator

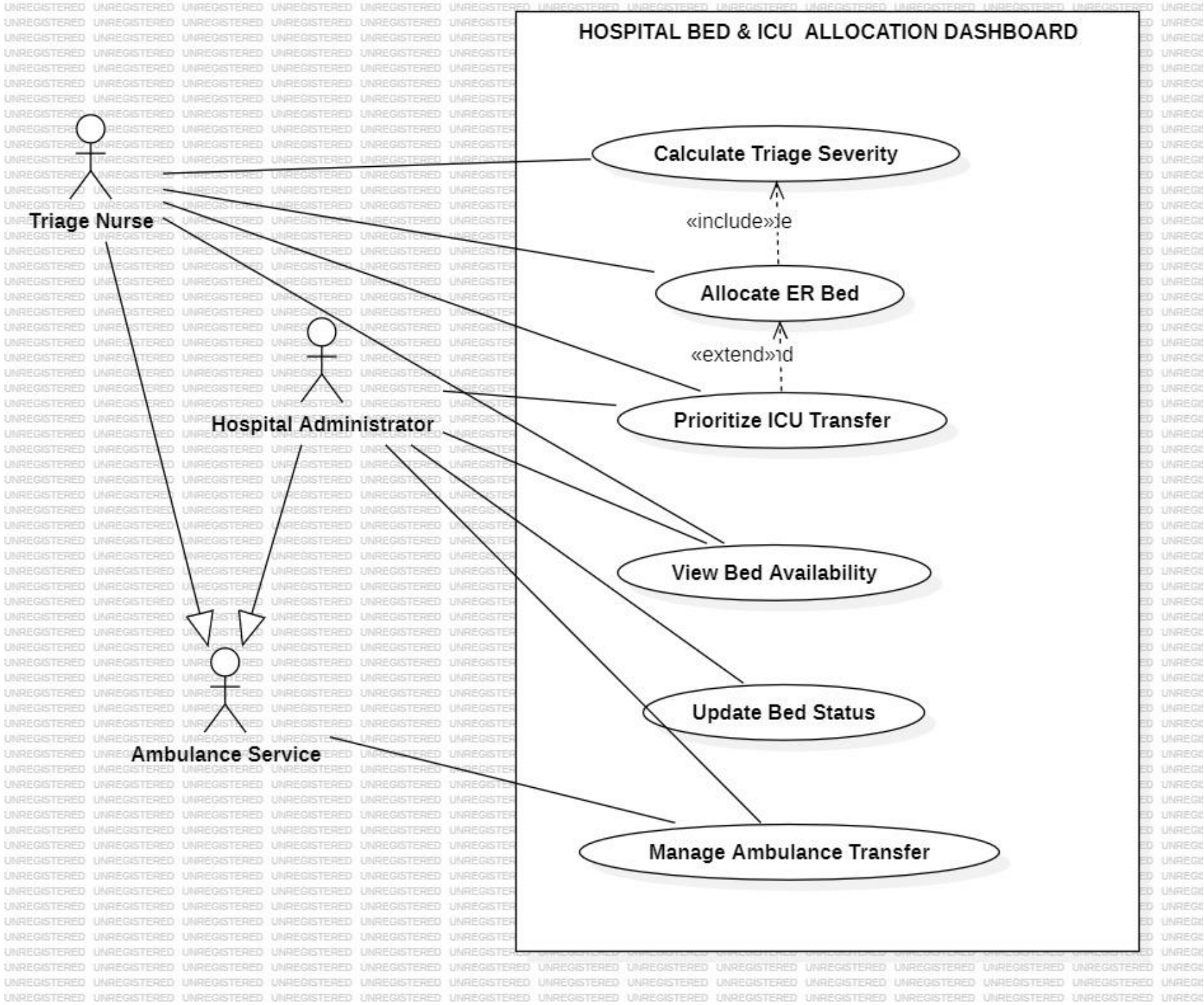
1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall compute triage severity scores from intake vitals and dynamically order the ER bed allocation waitlist.	High	**Pass:** A higher-acuity patient automatically moves ahead of a lower-acuity patient. **Fail:** The queue follows FIFO and ignores a critical emergency	Ensures critically ill patients receive priority for bed allocation	Requirement is clear and testable. Acceptance criteria clearly defines priority behavior.
FR-002	Functional	The system shall display the current status of hospital beds, including Occupied, Cleaning, and Available.	High	Pass: When a bed status changes, the new status is displayed on the dashboard. Fail: The dashboard continues showing the previous status	Helps staff identify available beds quickly.	Requirement clearly defines the required bed statuses. Status change can be verified on the dashboard.
FR-003	Functional	The system shall allow the Triage Nurse to assign an available bed to a prioritized ER patient.	High	Pass: Selecting an available bed assigns it to the selected patient and changes the bed status to Occupied. Fail: The patient is assigned without changing the bed status.	Supports efficient emergency bed allocation.	Requirement clearly identifies the Triage Nurse and the expected bed allocation behavior.
FR-004	Functional	The system shall allow the Hospital Administrator to initiate and track inter-facility ambulance transfer requests for patients	Medium	**Pass:** A transfer request is created with patient, destination, ambulance, and transfer status details, and the	Supports coordination of patient transfers between healthcare facilities.	Requirement clearly defines the ambulance transfer tracking information

		requiring transfer.		request status can be viewed by the Hospital Administrator. **Fail:** The transfer request cannot be created or tracked.		and status..
FR-005	Functional	The system shall notify the Triage Nurse when an ICU bed becomes available for a patient requiring ICU transfer.	High	**Pass:** When a suitable ICU bed becomes available, the system generates an alert for the Triage Nurse within 1 second. **Fail:** No alert is generated or the alert is delayed beyond 1 second.	Reduces delays in transferring critical patients to ICU.	Requirement clearly defines when the ICU availability notification should be generated.
NFR-001	Nonfunctional	Bed status transitions (Occupied, Cleaning, Available) must propagate to all connected ER dashboard clients in under 1 second via WebSockets.	High	**Pass:** Benchmarking confirms the updated bed status reaches connected dashboard clients in less than 1 second under simulated peak load. **Fail:** The update takes 1 second or longer.	Provides near-real-time bed availability information.	Requirement is measurable because the maximum update time is explicitly specified as under 1 second.
NFR-002	Nonfunctional	The system shall restrict access to patient and bed allocation information to authenticated and authorized users	High	**Pass:** Unauthorized users are denied access to protected patient and bed allocation information, while authorized users can access the information. **Fail:** An	Protects sensitive healthcare information and prevents unauthorized	Requirement clearly defines authentication and authorization as security controls for protected information.

				unauthorized user can access protected information.	access.	

2 Use-Case Diagram:


HOSPITAL BED & ICU ALLOCATION DASHBOARD
Triage Nurse
Hospital Administrator
Ambulance Service
Calculate Triage Severity
«include»le
Allocate ER Bed
«extend»id
Prioritize ICU Transfer
View Bed Availability
Update Bed Status
Manage Ambulance Transfer

Use-Case Flow

USE-CASE FLOW SPECIFICATION

Use Case: Allocate ER Bed

Use Case ID: UC-03

Primary Actor: Triage Nurse

Preconditions

1. The Triage Nurse is authenticated in the system.

2. Patient triage information has been entered.

3. At least one hospital bed is available.

4. The bed allocation dashboard is operational.

Postconditions

1. The patient is assigned to an appropriate available bed.

2. The selected bed status changes to Occupied.

3. The allocation information is recorded in the system.

4. The updated bed status is visible on connected dashboards.

Main Success Scenario

1. Triage Nurse opens the Bed & ICU Allocation Dashboard.
2. System displays the current ER bed availability and patient waitlist.
3. Triage Nurse selects a patient from the ER waitlist.
4. System calculates the patient's triage severity score from the available intake vitals.
5. System prioritizes the patient according to the severity score.
6. System displays suitable available beds.
7. Triage Nurse selects an appropriate available bed.
8. System assigns the bed to the patient.

9. System changes the bed status from Available to Occupied.

10. System updates the connected dashboard clients.

11. System confirms successful bed allocation.

12. Use case ends successfully.

Alternate Flow — No Suitable Bed Available

a. System determines that no suitable bed is currently available.

a1. System displays "No suitable bed available."

a2. System keeps the patient in the prioritized waitlist.

a3. System continues monitoring bed availability.

a4. When a suitable bed becomes available, the system notifies the Triage Nurse.

a5. Triage Nurse resumes the bed allocation process.